

Router Configuration

Booting up Process

- ❖ POST (Power on self test)
- ❖ Locate and load IOS
- ❖ Locate and run device configuration file



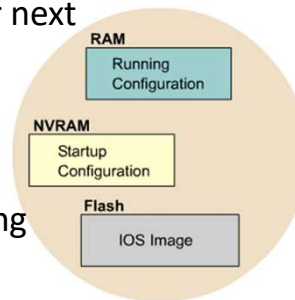
Fig . 3

ROUTER SOFTWARE OVERVIEW

- Routers depend on software for their operation.
- Two types of software required are operating system and configuration
- Software components stored in memory as a separate file.
- These files stored in different types of memory.

ROUTER CONFIGURATION

- Two types of configuration
- Startup configuration
 - In NVRAM
 - How the router will operate after next reload
- Running configuration
 - In RAM
 - How the router is currently running



IFS

- Version 12 of the IOS, a single interface to all the file systems
- Referred to as the Cisco IOS File System (IFS).
- Provides a single method to perform all the file system management
- Include flash memory file systems, the network file systems (TFTP and FTP), and reading or writing data (such as NVRAM, the running configuration, ROM).
- Uses a common set of prefixes to specify file system devices.

TFTP SERVER

- Virtual server
- Even make your router as a tftp server
 - Router# config t
 - Router(config)# tftp-server flash
- In Cisco router, the active configuration is in RAM and the default location for the startup configuration is NVRAM.
- One of these backup copies of the configuration can be stored on a TFTP server

Managing IOS images using TFTP

- When a router first arrives, the IOS should be backed up.
- The IOS backup can be initiated from the privileged EXEC mode with the copy flash tftp command.
- The router will prompt the user to enter the IP address of the TFTP server.

Using Router Diagnostic Commands

- Cisco routers provide numerous integrated commands to assist in monitoring and troubleshooting internetwork.
- Basic categories of these commands
 - **Show** : help monitor installation behavior and normal network behavior, as well as isolate problem areas.
 - **Debug**: Assist in the isolation of protocol and configuration problems.
 - **Ping**: Help to determine connectivity between devices on your network.
 - **Trace**: A method of determining the route by which packets reach their destination from one device to another.

Introduction to control messages

- The Internet Control Message Protocol (ICMP) is an integral part of the TCP/IP protocol suite.
- They are used to inform hosts of conditions such as network congestion or the existence of a better gateway to a remote network.
- Like all ICMP messages, ICMP control messages are encapsulated within an IP datagram.
- ICMP uses IP datagrams in order to traverse multiple networks.
- Multiple types of control messages are used by ICMP.

Address mask requirements

- If a host does not know the subnet mask, it may send an address mask request to the local router.
- If the address of the router is known, this request may be sent directly to the router. Otherwise, the request will be broadcast.
- When the router receives the request, it will respond with an address mask reply.

Router discovery message

- When a host on the network boots, and the host has not been manually configured with a default gateway, it can learn of available routers through the process of router discovery.
- This process begins with the host sending a router solicitation message to all routers, using the multicast address 224.0.0.2 as the destination address.
- When a router that supports the discovery process receives the router discovery message, a router advertisement is sent in return.

Router solicitation message

- A host generates an ICMP router solicitation message in response to a missing default gateway.
- This message is sent via multicast and it is the first step in the router discovery process.
- A local router will respond with a router advertisement identifying the default gateway for the local host.

Congestion and flow control messages

- Dropped packets occur when there is too much congestion on a network.
- ICMP source-quench messages are used to reduce the amount of data lost.
- The source-quench message asks senders to reduce the rate at which they are transmitting packets.
- Most Cisco routers do not send source-quench messages by default, because the source-quench message may itself add to the network congestion.